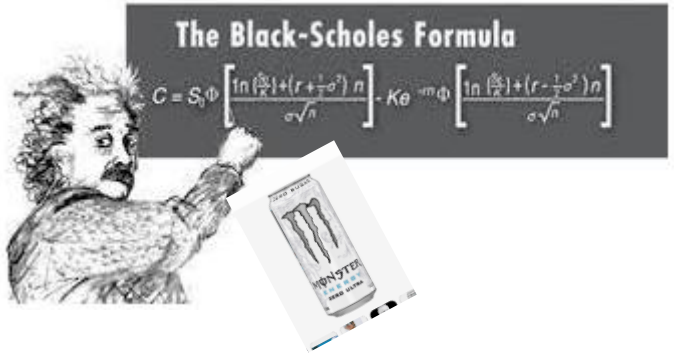
Using Geometric Brownian Motion and Black-Scholes to model European Call options

STAT 482 RANDOM PROCESSES W/ DR. OLGA

JOHN NAUGHTON 12/2/24



AKA the 3rd of 4 stock market presentations today

What is a European Call Option? From Simple to Complex

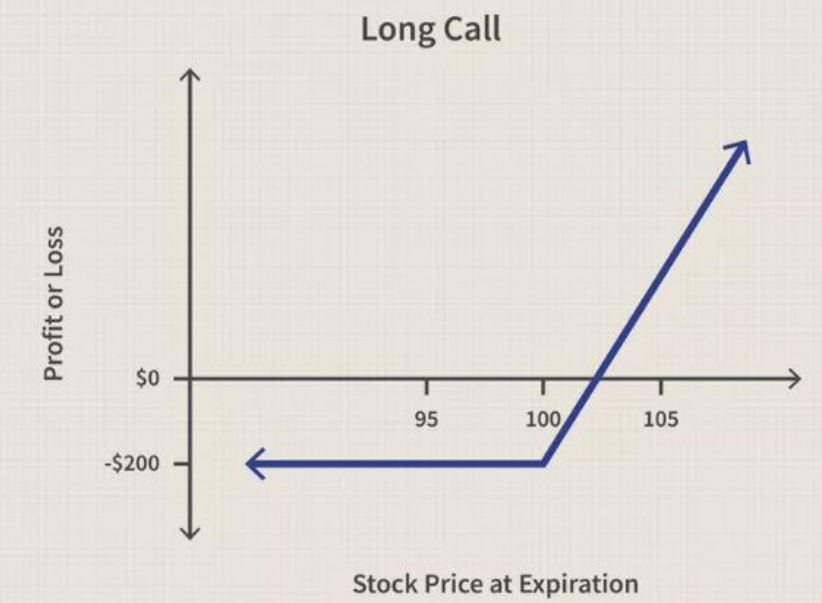
In the economy, money can be exchanged for goods and services

People form corporations to take advantage of economies of scale, and these corporations issue stock to raise funds for investment

Stocks collect dividend payments and are bought/sold on the stock market. Speculators trade in financial instruments that increase leverage and exposure to the movement of these assets.

The simplest of these derivative instruments are calls and puts: The buyer of a European call agrees to buy a stock for a certain price at a certain time in the future, hoping the price of the actual stock will increase and they can make a profit, while the seller of a European call will hope for the opposite. An “option” means that the buyer of a call or put(the opposite) is shielded from the downside risk, but in exchange they pay a premium to the option seller.

A new application for Geometric Brownian Motion



- 1973 Fisher Black and Myron Scholes elaborated on Geometric Brownian motion with the Black-Scholes model, which revolutionized finance by providing a approach to derivative pricing, leading to the development of the options market. It operates under the following assumptions:
- The asset price follows a lognormal distribution and evolves according to a geometric Brownian motion.
- Markets are efficient, with no arbitrage opportunities. There are no transaction costs or taxes. The risk-free rate and volatility are constant.
- The option can only be exercised at expiration (a European option. Conversely an American Option can be exercised at any time and therefore requires a more complex model.)

Black-Scholes Theory

We let C be the price of our option. Our assumption that arbitrage opportunities are zero gives us

$$\mathbb{E}\left[e^{-r\,t}\,(X(t)-K)^{+}\;-\;C\right]=0,\;\text{or}\;\;C\;=\;e^{-r\,t}\,\mathbb{E}\left[(X(t)-K)^{+}\right]$$

↑ Don't forget Merton!

X(t) is our stock price, modeled by geometric Brownian motion with drift parameter μ and volatility σ .

Using the independence and stationarity of increments of the Brownian motion

$$\mathbb{E}\left[e^{-r\,t}\,X(t)\,|\,X(u),\,0\le u\le s\right]=e^{-r\,t}\,X(0)\,\mathbb{E}\left[e^{\mu\,t+\sigma\,B(t)}\,|\,B(u),\,0\le u\le s\right]$$

We can simplify our option price function to

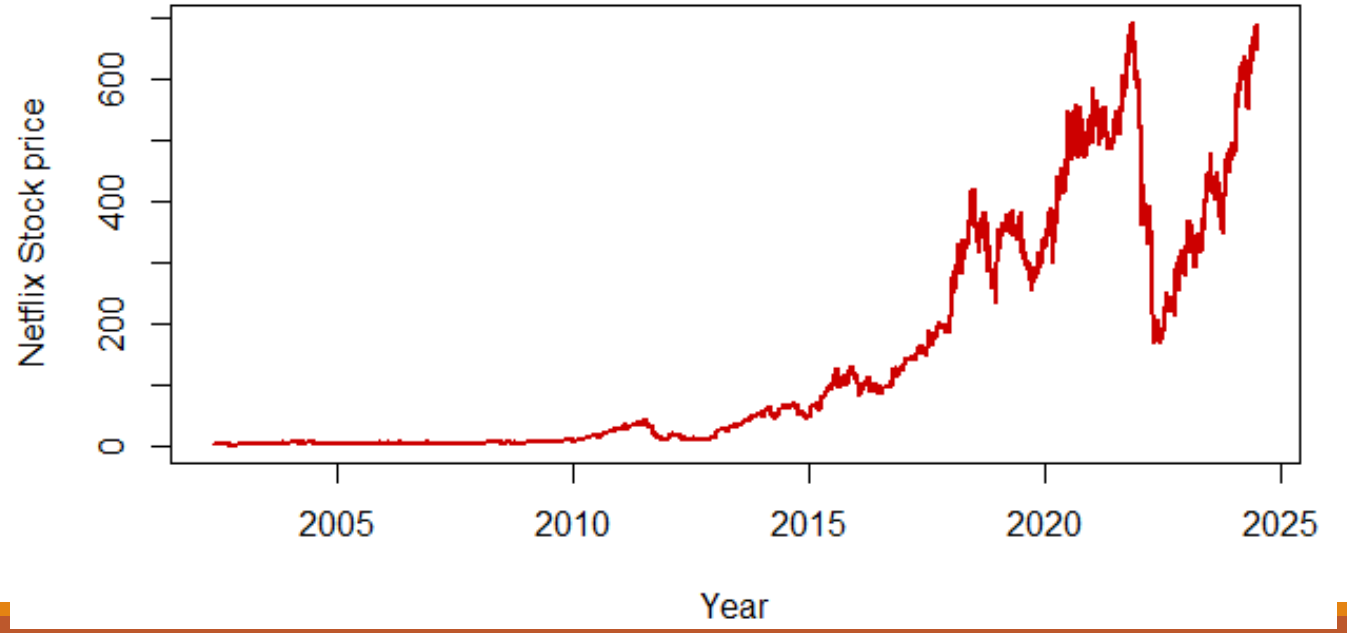
$$C=X(0)\,\Phi\left(A\;+\;\sigma\,\sqrt{t}\right)\;-\;e^{-r\,t}\,K\,\Phi(A)$$ where $$A\;=\;\frac{\mu\,t\;-\;\ln(K/X(0))}{\sigma\,\sqrt{t}}$$ and $$r\;=\;\mu\;+\;\frac{\sigma^2}{2}$$

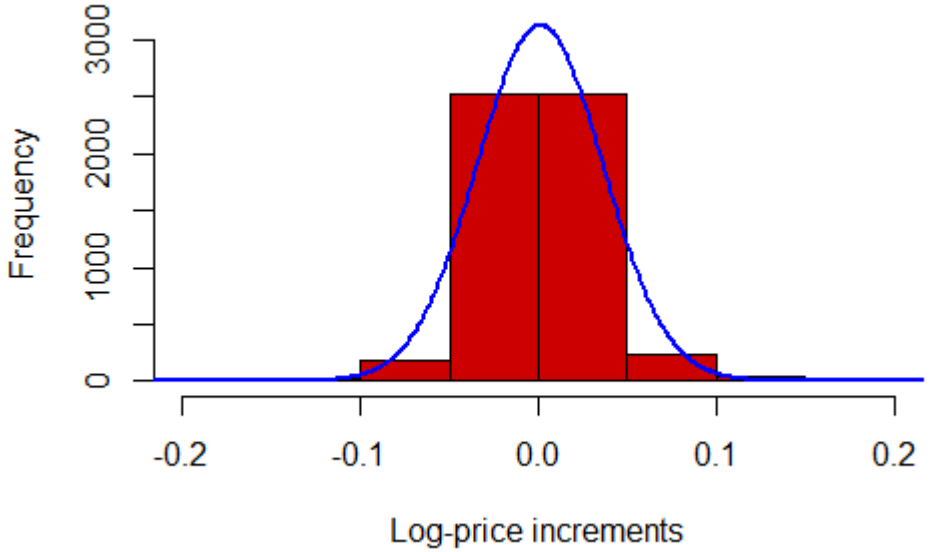
Which means C is now a function of μ, σ, current stock price X(0), time until strike date t, and option strike price K .

Data Set

Date	Open	High	Low	Close	Volume
5/23/2002	1.1564290	1.2428569	1.1457140	1.1964290	1.05E+08
5/24/2002	1.2142859	1.2250000	1.1971429	1.2100000	11104800
5/28/2002	1.2135709	1.2321430	1.1571429	1.1571429	6609400
5/29/2002	1.1642860	1.1642860	1.0857139	1.1035710	6757800
5/30/2002	1.1078569	1.1078569	1.0714290	1.0714290	10154200

We decide to model the Netflix stock and option market, using a dataset on Kaggle that contains close prices for every market day starting with Netflix's IPO on May 23rd 2002 at **1.19 dollars** per share to July 15th 2024 at **656.45 dollars** per share for 5573 observations total.



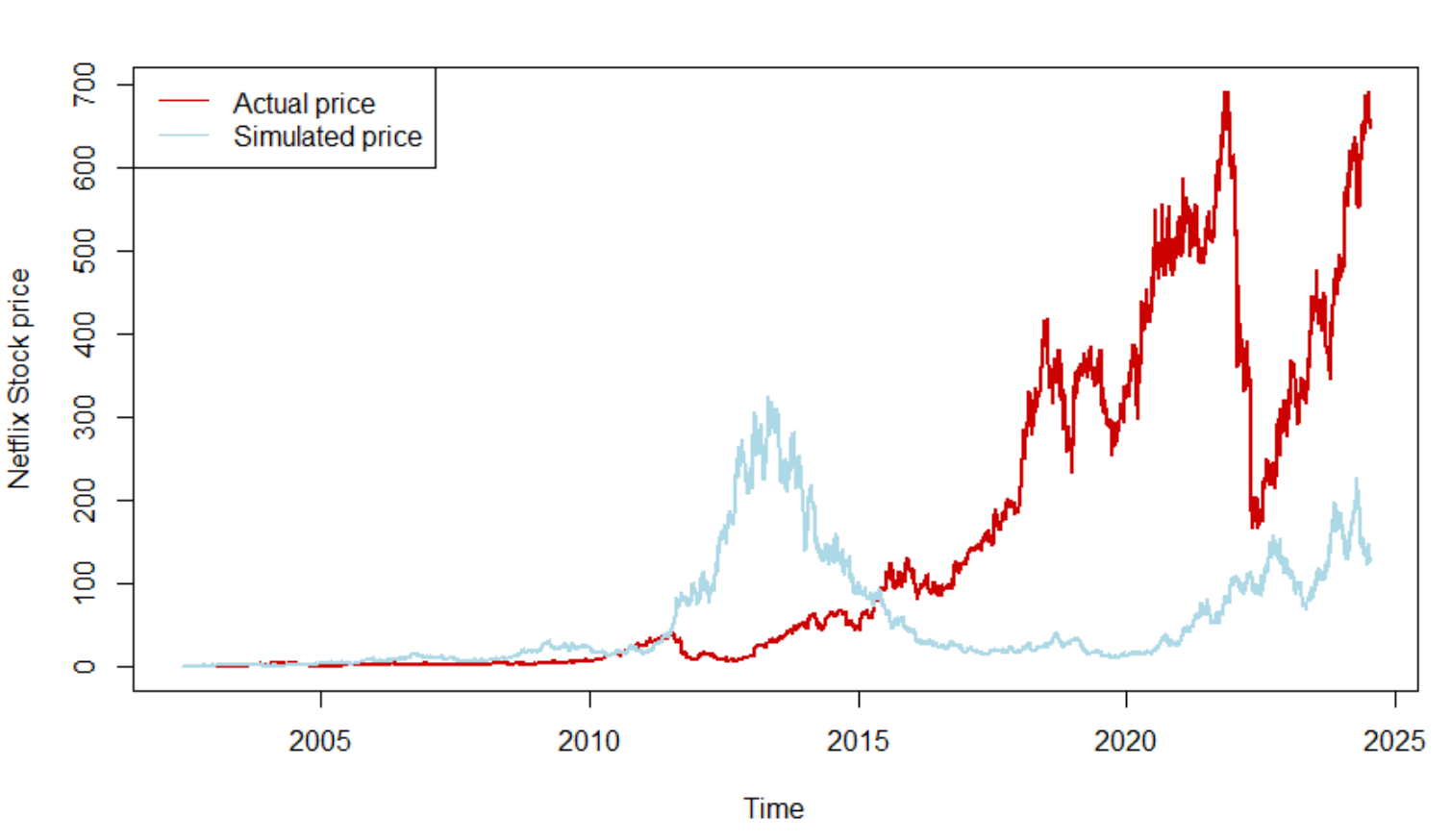


From the histogram we conclude the distribution of log-price increments is normal.

```r
stock.data<- read.csv(file="C:/Users/jknau/OneDrive/Desktop/482 Random Processes/NFL
str(stock.data$Date)
stock.data$Date <- as.Date(stock.data$Date, format="%m/%d/%Y")
str(stock.data$Date)
Date<- as.POSIXct(stock.data$Date)

price<- stock.data$Close
#plotting stock price against date
plot(Date, price, type="l", lwd=2, cex=0, col="red3",
     xlab="Year", ylab="Netflix Stock price")

#calculating increments of log-price
log.inc<- c()
price1<- price[-1]
price1.lag<- head(price, -1)
log.ratio<- log(price1/price1.lag)
#plotting histogram
library(rcompanion)

plotNormalHistogram(log.ratio, xlab="Log-price increments",
                    col="red3", xlim=c(-.2, .2))
```

Code

Estimating drift and volatility parameters and plotting a single trajectory




```
> print(mu.hat<- mean(log.ratio))
[1] 0.001132
> print(sigma.hat<- sd(log.ratio))
[1] 0.03554163
```

```
#estimating parameters
print(mu.hat<- mean(log.ratio))
print(sigma.hat<- sd(log.ratio))

#specifying Brownian motion as vector
BM<- c()
#specifying initial value
BM[1]<- 0
#specifying seed
set.seed(34235457)
#simulating Brownian motion with drift and volatility
for (i in 2:5573)
  BM[i]<- mu.hat+BM[i-1] + sigma.hat*rnorm(1)
#computing values for geometric Brownian motion
GBM<-price[1]*exp(BM)
#plotting actual and simulated trajectories
plot(Date, price, type="l", lty=1, lwd=2, col="red3",
     xlab="Time", ylab="Netflix Stock price")
lines(Date, GBM, lwd=2, col="lightblue")
legend("topleft", c("Actual price", "Simulated price"),
       lty=1, col=c("red3", "lightblue"))
```

Making a prediction with Black-Scholes

- The current price (12/3/24) of Netflix stock is **902.16 dollars/share**, but let's say we're very optimistic about the company and want to invest. In addition, we want to leverage our position by trading in a financial derivative while also being shielded from the downside risk.
- We buy a European call option that gives us the right to buy a share of Netflix stock for **950 dollars/share** (the strike price) in **30 days**. What should we pay for this option/ what does the market price this option at?

NETFLIX

Calculations

μ = .001132

σ = .03554163

X(0)= 902.17

t= 30

K = 950

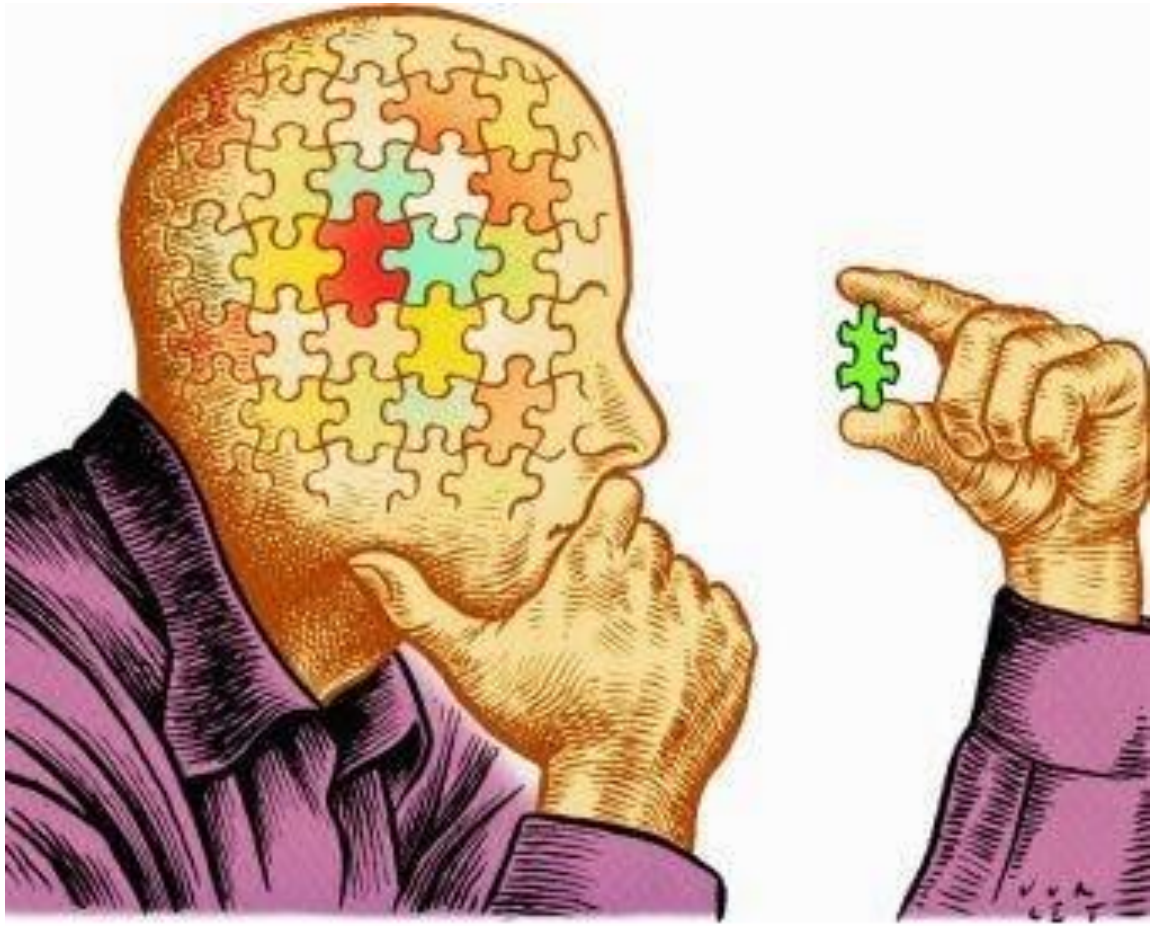
$$A = \frac{\mu t - \ln(K/X(0))}{\sigma\sqrt{t}} = -.0909227$$

$$r = \mu + \frac{\sigma^2}{2} = .0017636$$

$$C = X(0)\,\Phi\left(A + \sigma\sqrt{t}\right) - e^{-rt}\,K\,\Phi(A)$$

= 902.17 ϕ (.10373) – 901.0439 ϕ (-.0909227)

= 70.47 dollars for European call option expiring in 30 days at $ 950 .



Closing thoughts

- Although the pricing models in the finance world have become incredibly complex, the fundamentals of Black Scholes still stand.
- The actual price for our call option on sites like Robinhood were closer to 15-20 dollars, I suspect that using the entire 22-year history of Netflix's stock price is possibly less accurate than a more relevant time window because it overestimates structural volatility.

Thank you everyone!

Especially Dr. Olga! Good luck with the final push this semester. I am graduating, feel free to add me, John Naughton, on Linkedin.